
Embedded Systems International

Project Statement of Work (PD3)

Team Identifier: SM 3-1

Team Name (optional):

Team Member Names: Rose Reinhart, Dylan Kramer

Submit your document as a PDF file in Canvas under the corresponding project assignment.

One of your section's lab TAs must approve of this Statement of Work by adding a comment in the Canvas assignment. The team is responsible for requesting approval promptly after submission.

Refer to the Project Requirements document before completing this Statement of Work (SOW). A statement of work is a focused concise proposal and agreement that describes work to be done. Teams should complete and submit this SOW form, which represents several parts of a statement of work, including a plan for what you are doing and how. The SOW defines the scope of your project and the approach you are taking to deliver on the goals.

Problem Statement

First, has your team reached consensus on the autonomous vehicle (AV) application you will use as the context or story for your project? All projects, regardless of application, will need to meet the same basic requirements and will be recognized for innovative features.

To define your problem, think about one or more users and their needs. Write at least one Point of View (POV) statement for your application. Follow Steps 2 and 3 in the following guide (also in Canvas as a PDF document):

[Define and Frame Your Design Challenge](#) (links to IDF webpage)

URL:

<https://www.interaction-design.org/literature/article/define-and-frame-your-design-challenge-by-creati>

[ng-your-point-of-view-and-ask-how-might-we](#)

See the Lab Project page in Canvas for the PDF file for the webpage.

Next, think about and write a few sentences that give a high-level summary of the broad mission goals for your AV application. For example:

The purpose of the AV is to provide

The AV will be capable of doing

The mission goals and user needs establish the purpose of the project and why you are working on it. Now you should translate these into a more detailed problem statement that provides a specific, concise, clear and thorough description of the context for the problem, an explanation of user needs that will be addressed, and an outline of your proposed technical approach to solving the problem.

Problem Statement

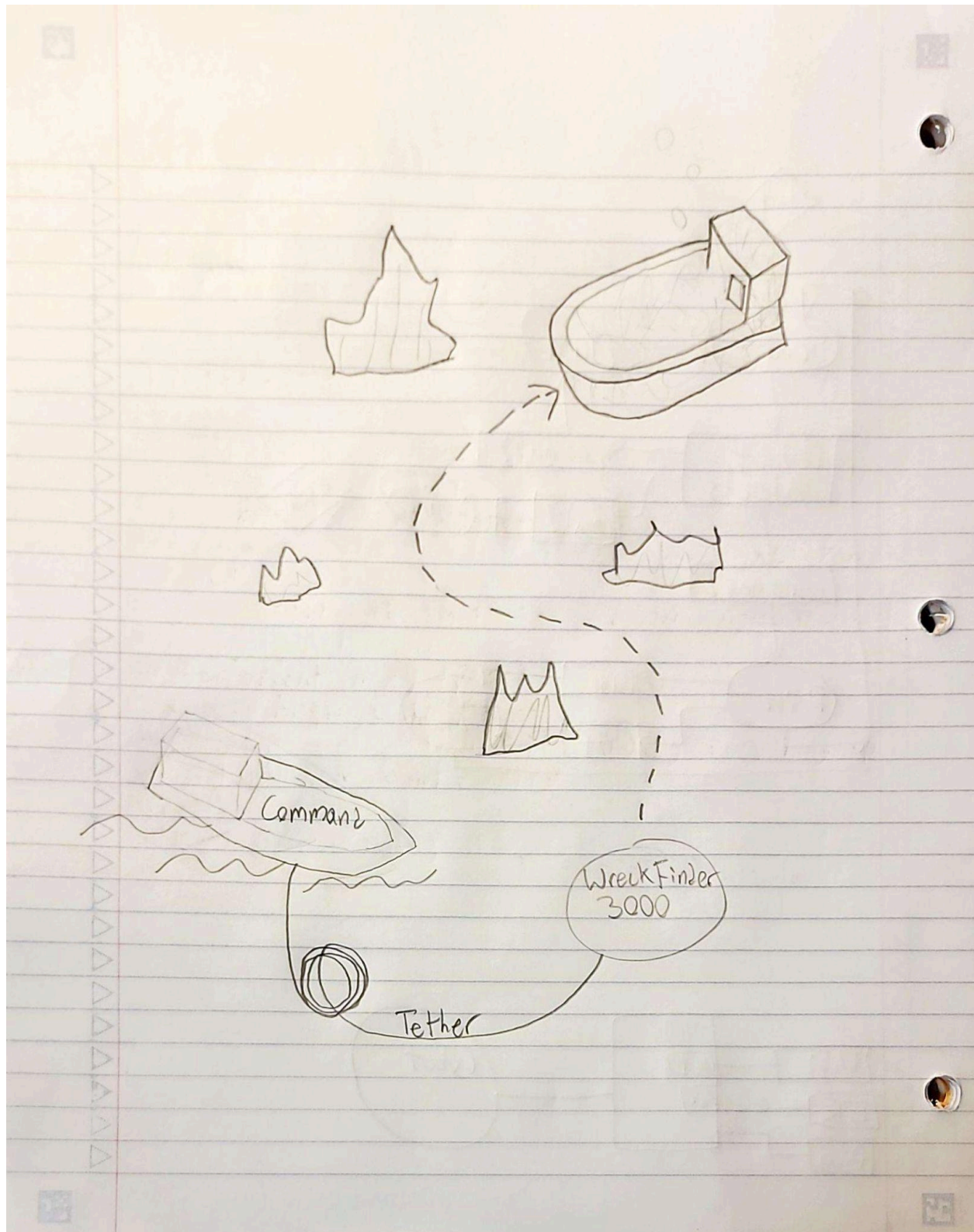
POV statement: Mrs. Johnson, a dear friend, is struggling with the loss of her husband. He was out fishing with friends on the Pacific Ocean, when his boat went down. She has not been given answers, and years have gone by. Wanting to help, the WreckFinder 3000 was created. This engineering masterpiece can be deployed in under a minute, and has IR and ultrasound sensors to scan the ocean floor looking for the wreck. This robot can give Mrs. Johnson the clarity and answers she's wanted for years.

High-level mission goals: This project is inspired by Mrs. Johnsons tragedy, alongside many others. This project also is meant to protect search and rescue teams from having to dive in dangerous waters/conditions to search, when a robot could take the risk for them. On top of that, high-pressure diving has a depth limit, where this robot has a much deeper limit. The idea is to leverage the advanced sensors and capabilities of this robot to advance search and rescue operations nationwide. Not only can this robot be used to find debris in shipwrecks, but to find almost anything in deep waters.

Short description of the problem:

Search and rescue operations are dangerous. Dark, deep, murky and fast-moving waters present real risks for dive teams. Rose and I set out to create the newest engineering masterpiece, the WreckFinder 3000. This robot uses advanced sensors to traverse the ocean floor to find debris. It reports its findings in real-time back to the search and rescue boat through a serial communication diving line (wifi to PuTTY in our simulation). The robot will send information about where debris is in relation to it, allowing the operator in the command boat to send commands back on where to move. This robot should significantly decrease search and rescue time, increase the safety of the team, and also enhance the capabilities of the search. The WreckFinder 3000 is about to disrupt the industry.

In addition to writing a paragraph about the problem, you are to draw a **problem sketch: a one-page sketch illustrating your solution with a user context (big picture view)**. This should show the scope of your work in relation to one or more user needs. Refer to the sample project sketches.



Design Approach

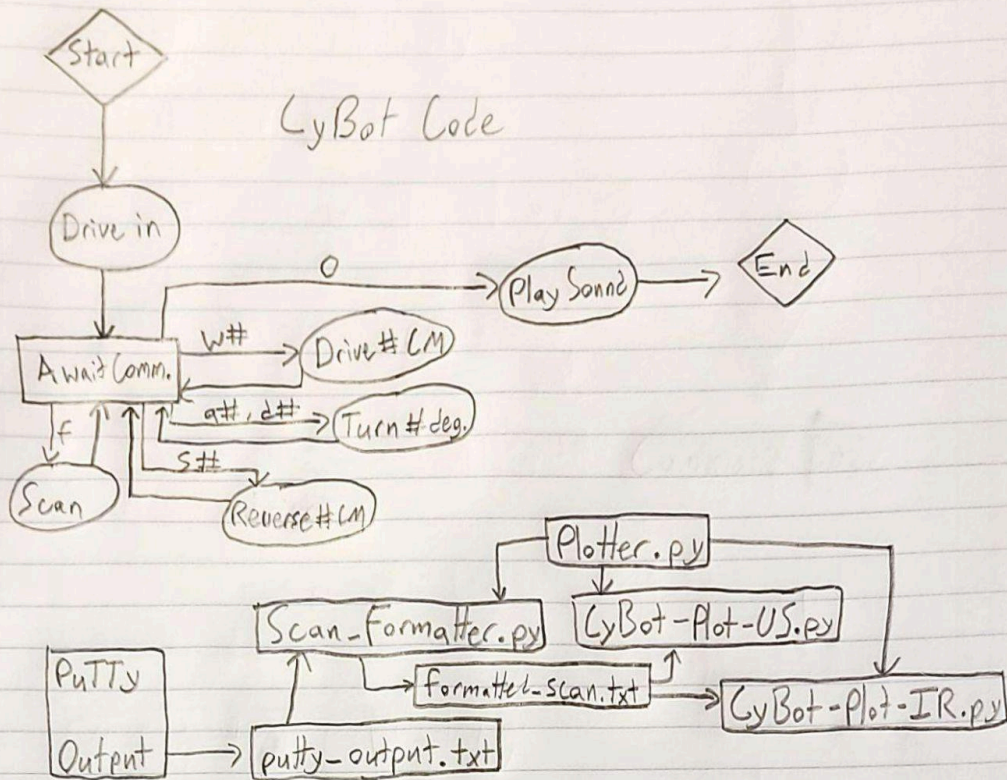
Next, consider your AV application in relation to the project requirements and the five categories by which it will be evaluated.

- 1) Functionality in relation to the AV application mission goals and user needs
- 2) Mapping of functional requirements to platform components and capabilities
- 3) Elements of the test field
- 4) Serious incident penalties
- 5) Feature bonuses

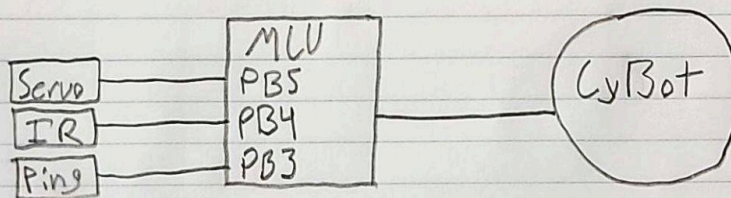
In this section, you will identify and describe how you will design your application for each of these categories.

To complete the tables below, your team may want to use ideation tools, such as Lotus Blossom. Refer to the Lab Project page in Canvas and see the project ideas guidelines and Lotus Blossom worksheet as needed.

In addition to completing the tables below, you are to draw a **technical system sketch: a one-page sketch depicting a high-level technical system diagram of your proposed solution, such as a block diagram or dataflow diagram**. This should show both hardware and software modules.



Command Computer Code



The tables below are your initial proposal, and you may update these before your demonstration.

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Functionality

Describe each of the basic functionalities required for the project in terms of your AV application. **The functionality must be specific to the problem and user(s).** Several functional statements are given in the example functional description for the Mars rover application in the Project Requirements document.

Basic Functionality	Mapping to AV Application
Cybot Communication	The CyBot will be using advanced wireless technology to communicate back to the command boat (Wifi from CyBOT to PuTTY in our case).
Cybot Movement	The CyBot will use its propeller jets (wheels) to traverse the floor of the ocean, and its ultrasound and IR sensor to detect debris in its path.
Object Detection	Object detection will use the ultrasound and IR sensor mounted to the front of the underwater robots servo motor, and send scan data back to the command boat to decide where to move.
Object Avoidance	The CyBot will use its propeller jets (wheels) to navigate around debris based on what the ultrasound and IR sensors find, and based off the commands the control boat sends wirelessly.
Boundary Adherence	The CyBot will use a 3D map that is drawn on its software by the search and rescue team (in our case it will use cliff sensors to detect lines on the side of the test field that simulate boundaries).
Arrival at Destination	When it arrives at the shipwreck it will mark it's location (make a sound) and send that information to the surface before stopping in the wreck (4-pillars arranged in a box-like fashion).
User Interface	The user interface will be proprietary mapping software that the robot directly communicates with (PuTTY in our simulation).

Base Station Control	The base station will charge the robot and temporarily turn it off. The base station is where the robot can receive updates as well.
Other Application Specific Functionality (may be novel features for bonus points)	
The robot will start outside the test field and drive into the test field before the search. Also, the robot, when it finds the shipwreck (4 small pillars arranged in a box-like fashion), it will drive inside of it and play a sound to prove it has found it.	

Mapping to Platform

Briefly describe how each of the basic platform components required for the project will be used in your AV application.

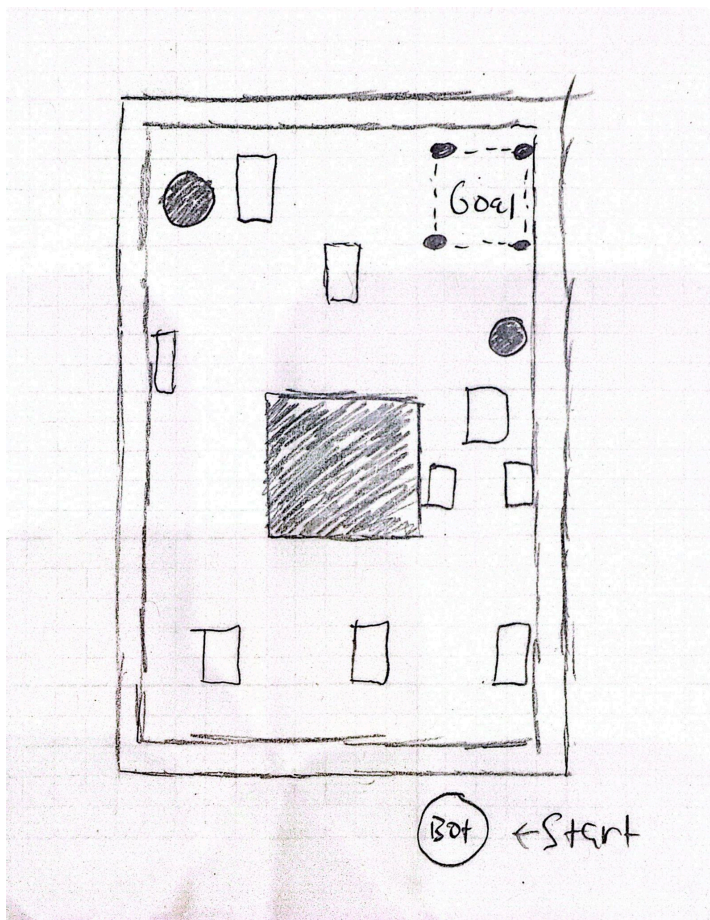
Basic Platform Components	Usage in AV Application
Open Interface and iRobot Sensors	The WreckFinder uses its Propeller Jets (wheels) to traverse the test field, and its PING sensor and IR sensor to accurately find debris at different distances.
Interrupts	Not required
ADC	The IR sensor feeds into the ADC in order for us to read it as input and calculate a distance value from it.
Input Capture	Our command terminal captures keyboard input that tells the robot what to do. For example, the operator can specify 'move forward 1 meter', or 'turn left 90 degrees'. These are done through commands such as w10 (move forward 1 meter), or a90 (turn left 90). This robot is controlled using W,A,S,D,F and O. W,A,S,D are standard game controls for vehicles, and F is used to initiate a 180 degree sensor scan, and O plays a sound.
PWM	The robot will use PWM to move a servo motor so it can scan from 0 to 180 degrees for more accurate scanning and object detection.
UART/WiFi	The robot will output it's scan data as well as it's object detection data using UART and Wifi.
Other Platform Components or Modes (may be novel features for bonus points)	
When the robot navigates into the shipwreck, the operator can play a sound by hitting 'O' and then 'enter' on the keyboard.	

Elements of the Test Field

Briefly describe a test field in the context of the real application (e.g., Martian terrain, city streets, etc.). Then state what each of the basic objects and other elements required for the test field represent in terms of the AV application. Draw and attach a **sketch of a possible simple test field for the lab.**

Test Field Description

The test field will consist of multiple different size debris scattered all over the place. In the test field there will be a group of 4 pillars in a box-like fashion that simulate going into a shipwreck. The gaps in between each of the 4 pillars will be large enough for the bot to pass through. Once the robot finds the shipwreck and enters, a sound will be played to demonstrate that the robot has found the wreckage.



Basic Objects and Other Elements	Mapping to AV Application Test Field
Tall objects (wide or composite)	Tall objects are big pieces of debris we want to avoid.
Short objects	Short objects are objects we may bump into as they are shorter than our sensors can detect, meaning that we need to back ourselves out of a potential trap once we hit one. Our robot is programmed to automatically detect a bump, backup, and alert the operator of the collision.
Holes	Holes are to be avoided at all cost, as they simulate a drop or ledge. Our robot uses cliff sensors to detect cliffs/drops and avoid them by immediately stopping the forward motion, then backing up slightly and notifying the operator of the cliff.
Pillars (thin tall object)	The shipwreck is simulated by 4 pillars arranged in a box-like fashion, and so if we see one, we need to make sure it's not the box, and if it is, drive inside of it.
Out of bounds	Out of bounds will be detected using the robots cliff sensors. If a cliff is detected AFTER the first time (driving into the test field), the robot will stop forward movement, backup slightly, then notify the operator.
Destination zone	The destination zone is directly in the middle of the 4 pillars that simulate the shipwreck.
Other Application Specific Elements (may be novel features for bonus points or incidents to avoid) The robot, when it finds the shipwreck (4 small pillars arranged in a box-like fashion), it will drive inside of it and play a sound to prove it has found it.	

Serious Incidents to Avoid and/or Novel Features (Optional)

You may have identified novel features in the tables above. Enter them in the table below and propose possible bonus points if demonstrated successfully. In addition, describe any additional serious incidents that might happen in your test field for your AV application.

Novel Features	Bonus Points
Time is of the utmost importance with search and rescue. Therefore, our robot will be done with its search in no more than 25 minutes (We will demo before prep week, allowing 30 minutes max). (10 pts) Another novel feature is the bot will start outside the test field, and drive into the test field before starting navigation. (5 pts) Our robot will play a sound once it drives into the shipwreck. (5 pts)	20

Serious Incidents	Deductions
Robot cannot drive outside the test field (-5)	-5
Robot cannot drive into the hole on the test field (-2)	-2
Robot cannot take more than 25 minutes (-2)	-2

Sketches

Attach the following sketches to your submission. These were noted above in red.

- Problem sketch
- Technical system sketch
- Test field sketch

Visual communication is helpful for sharing information. There are many ways to represent your information. The problem sketch could be anything from a cartoon (informal) to a UML use case diagram (more formal). The technical system sketch is similar to what has been used/shown in class and lab. The test field sketch probably needs no explanation.